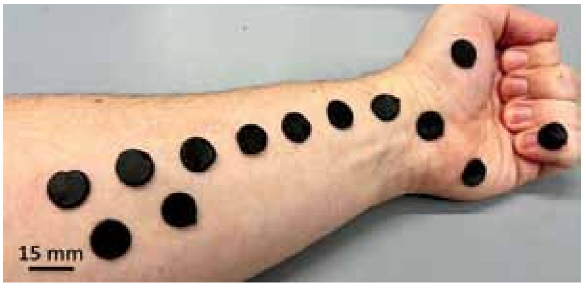
Play putty for flexible biometric sensors

Researchers at the University of Massachusetts Amherst have developed a low-cost alternative for monitoring brain, heart, muscle, and eye activity using homemade play putty called 'squishy circuits.' Made from common ingredients



Play putty for flexible biometric sensors (Credit: https://www.umass.edu)

like flour, water, salt, cream of tartar, and vegetable oil, the salt provides conductivity. Typically used as an educational tool, these circuits can now measure bioelectrical signals such as EEG, ECG, EOG, and EMG. They perform as well as commercial electrodes but cost just one cent per electrode. The putty is reusable, adaptable to skin contours, and offers an affordable, accessible option for bioelectrical measurements.

P-type semiconductors for displays

Korean researchers from ETRI have developed new P-type semiconductor materials and thin-film transistors, advancing semiconductor technology. They created a P-type selenium-tellurium (Se-Te) alloy transistor, which can be deposited at room temperature, improving manufacturing efficiency. Using a heterojunction structure with tellurium thin films, they developed a method to control N-type transistor threshold voltage. This enhances transistor mobility, on/off current ratios, and stability without needing a passivation layer. Their work is expected to drive innovations in high-refresh-rate displays, improving resolution and reducing power consumption. The team aims to commercialise these semiconductors for larger substrates and various circuits.

Streamlining 2D material twistronics

Thin 2D materials have potential for transistors, solar cells, and quantum computers. Researchers at Harvard University have developed a tool that controls electron density and twist angle in these materials, expanding research possibilities. They faced challenges with replicating twisted bilayer graphene at MIT due to the labour-intensive process. In response, they created MEGA2D, a micro-electromechanical system (MEMS)-based platform that simplifies the twisting of two material layers. This collaboration between the Yacoby and Mazur labs is adaptable for various materials, not just graphene. Tests on hexagonal boron nitride have revealed quasiparticles with properties useful for optical communications, suggesting this system could lead to breakthroughs in twistronics and assist further research in the field.

Ambient temperature energy harvesting

A team at Kyushu University's OPERA has developed an innovative organic thermoelectric device that can harvest energy from ambient temperature without needing a temperature gradient, unlike traditional thermoelectric devices.

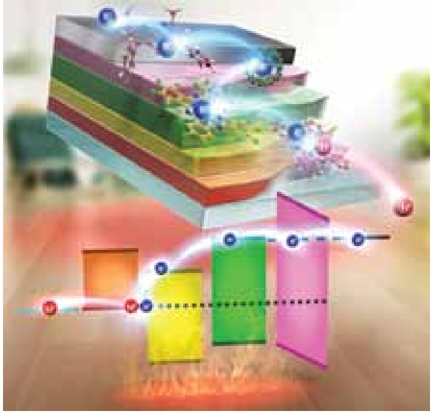


Illustration of the organic thermoelectric device (Credit: Kyushu University/Chihaya Adachi)

Conventional devices, such as those used in space exploration, rely on a temperature difference to generate electricity. The research team identified copper phthalocyanine (CuPc) and F16CuPc as key materials, enhancing performance with fullerenes and BCP. The device generated 384mV and 94nW/cm² at room temperature. This breakthrough shows the potential of organic materials for future energy technologies, with plans for further optimisation.

Ultrathin device tracks health markers

Researchers at the Korea Institute of Science and Technology (KIST) have developed a cutting-edge wireless device to monitor multiple biomarkers like glucose, lactate, and pH levels. This ultra-thin device, just 4µm thick, combines organic electrochemical transistor (OECT) sensors with inorganic micro-light-emitting diodes (µLEDs). The OECTs detect biomarkers by altering their current, which controls the µLEDs' light output for effective monitoring. The device demonstrated excellent transconductance (15mS) and mechanical stability in tests. Future improvements, such as integrating soft batteries or solar cells, could lead to a fully chipless, self-powered medical sensing system.